

CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY

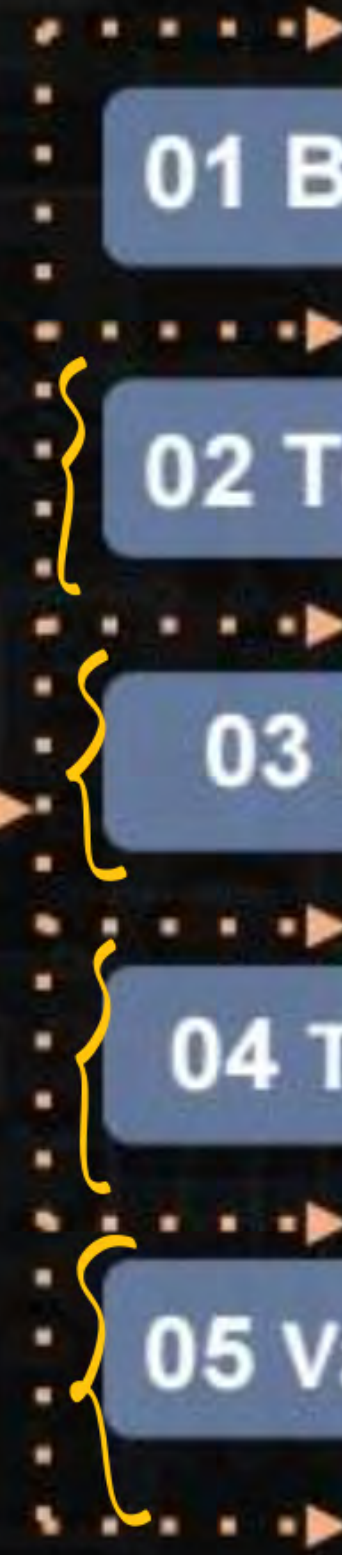


Lecture No. 3



By- SATISH YADAV SIR

TOPICS TO BE
COVERED



01 Basics of Functions

02 Terms in Functions

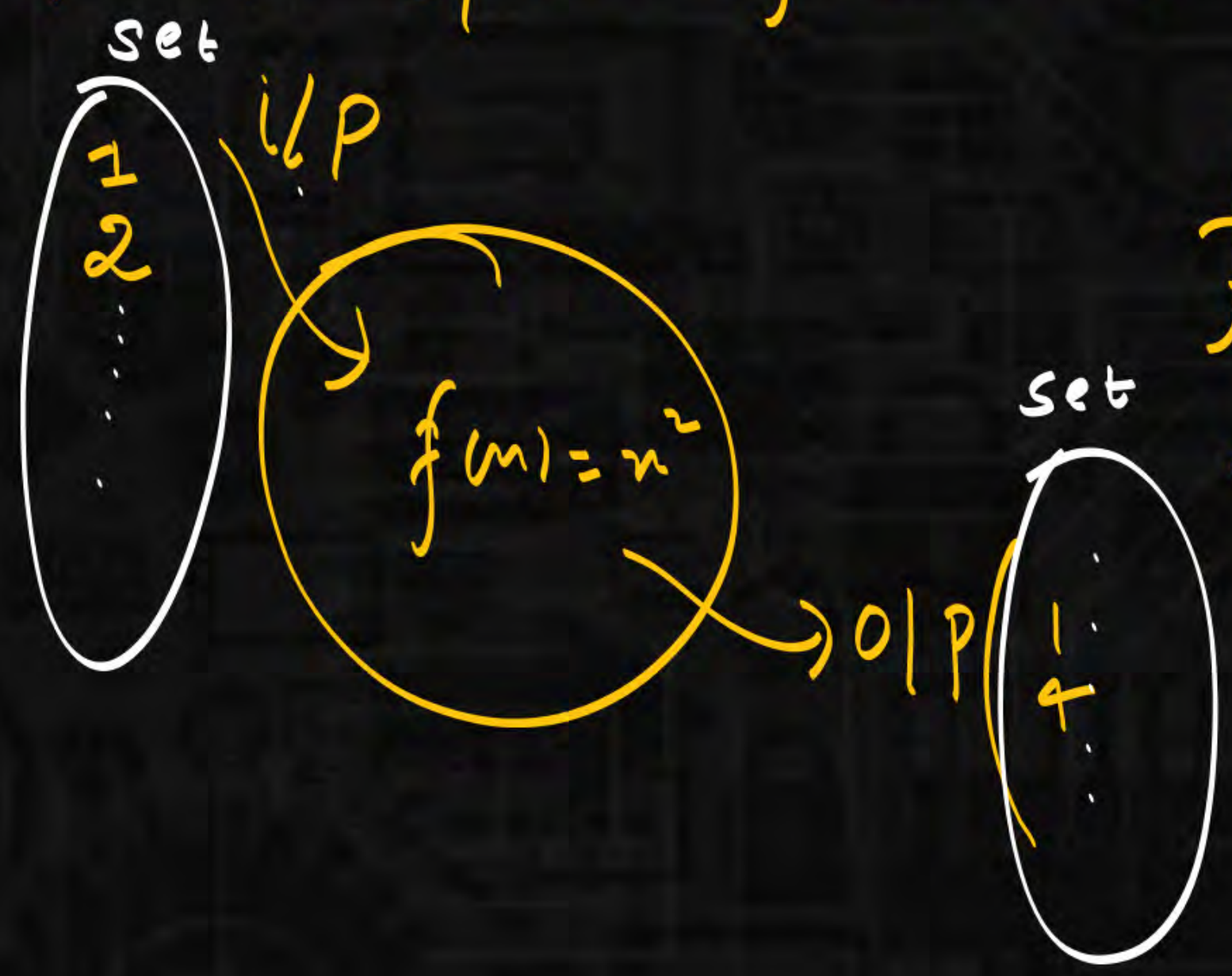
03 Number of Functions

04 Types of Functions

05 Various Examples in Functions

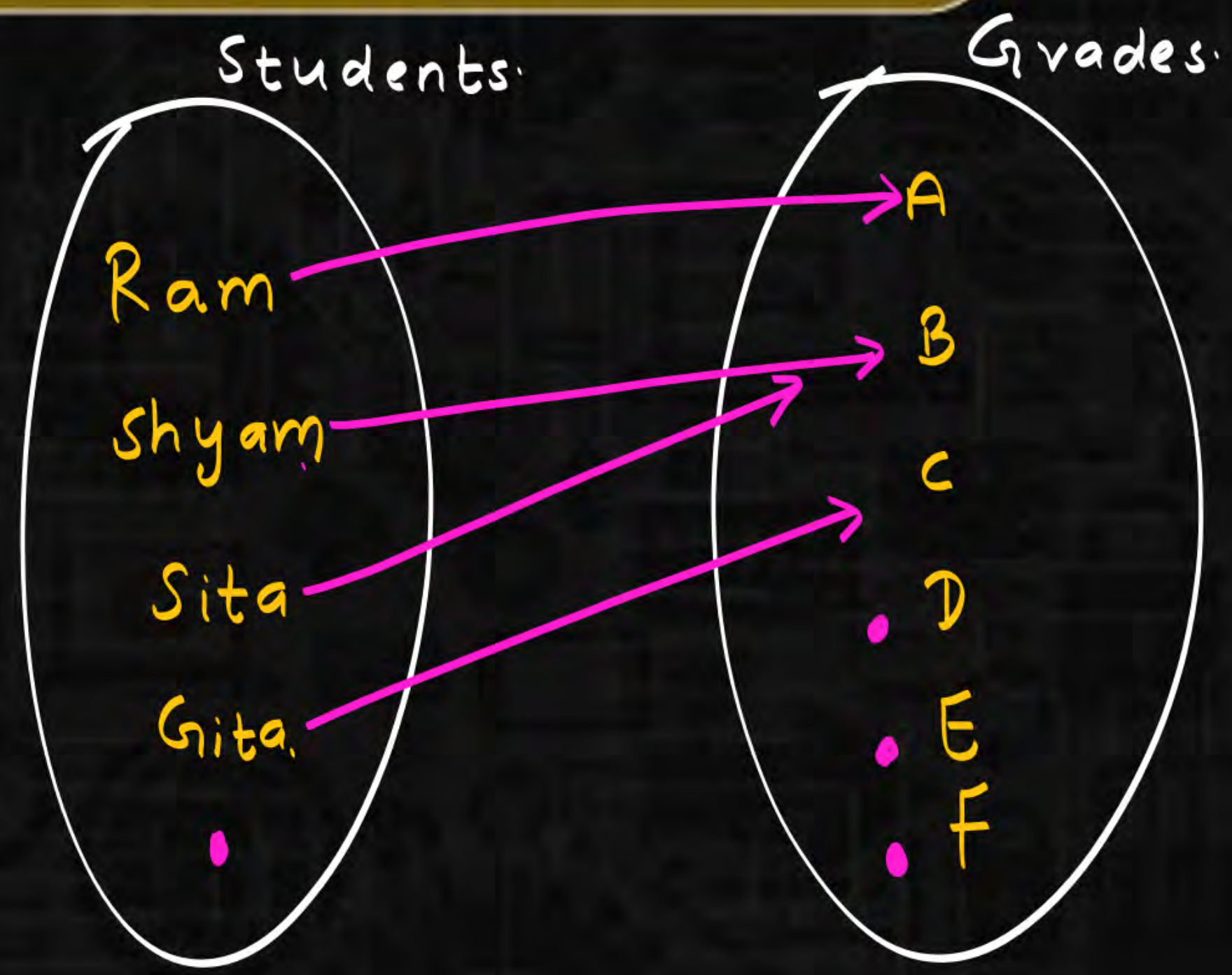
Functions

function/mapping/transformation



$$\begin{aligned} f(u) &= u^2 \\ f(u) &= e^u \end{aligned}$$

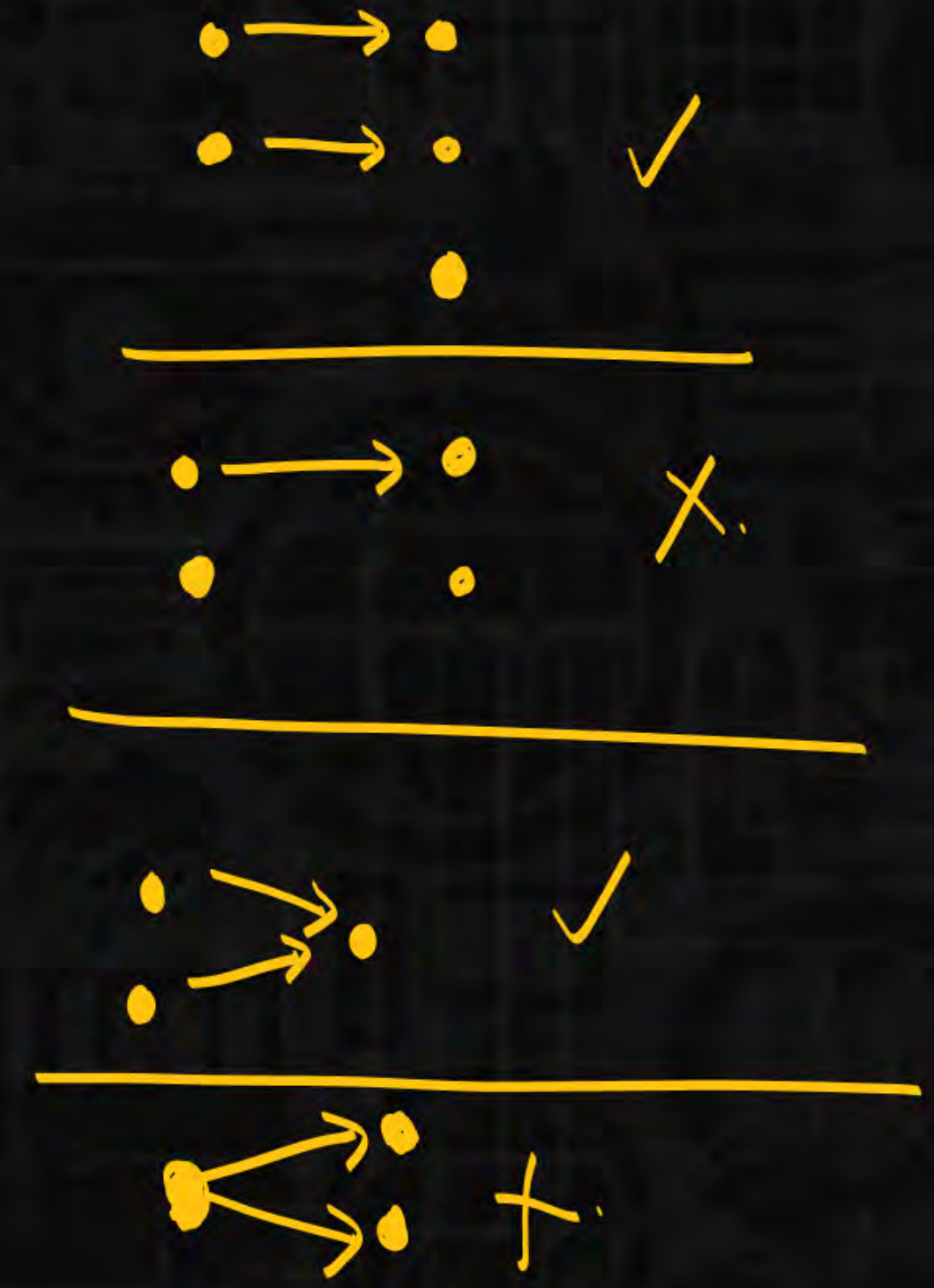
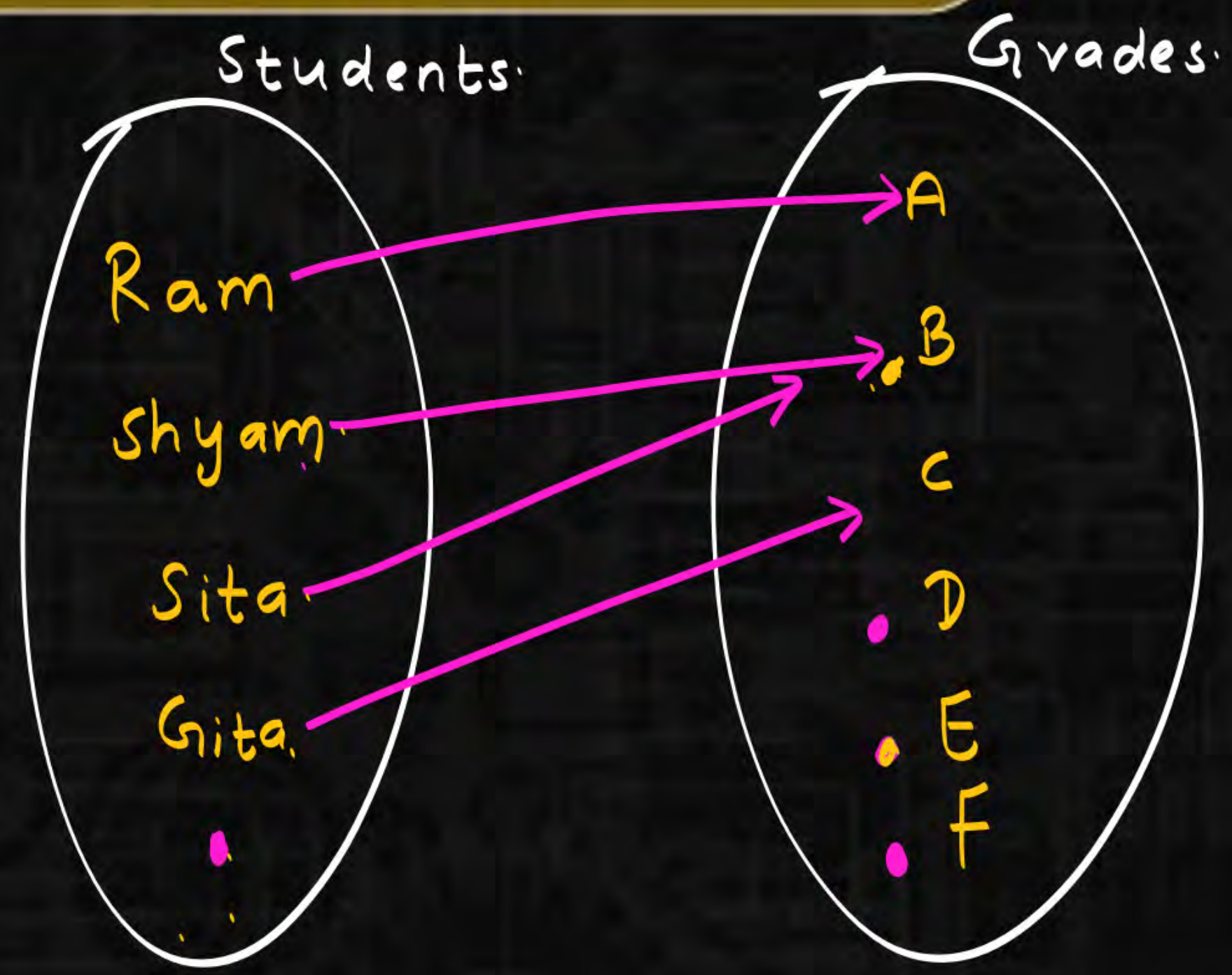
Functions



$$f : \text{Set 1} \rightarrow \text{Set 2}$$

$$f : \text{Students} \rightarrow \text{Grades}$$

Functions



Functions



Sem 1

Sita → 70
80

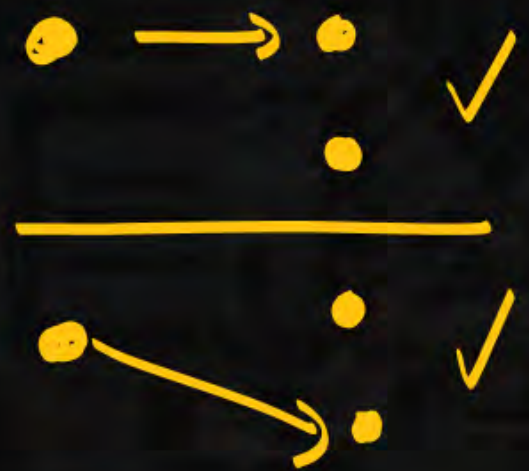
Sem 2

Sita → 70
80

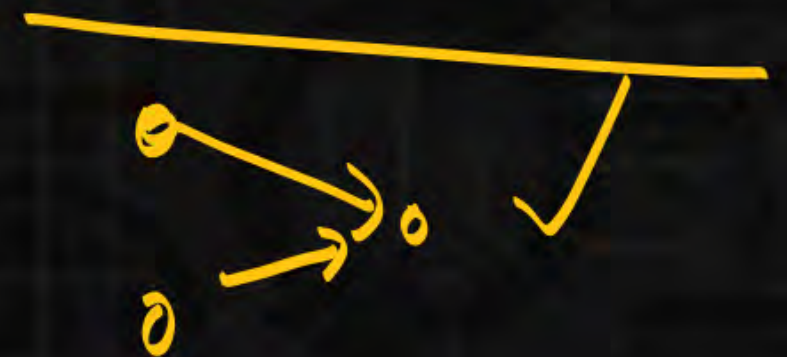
Sem 3

Sita → 70
80

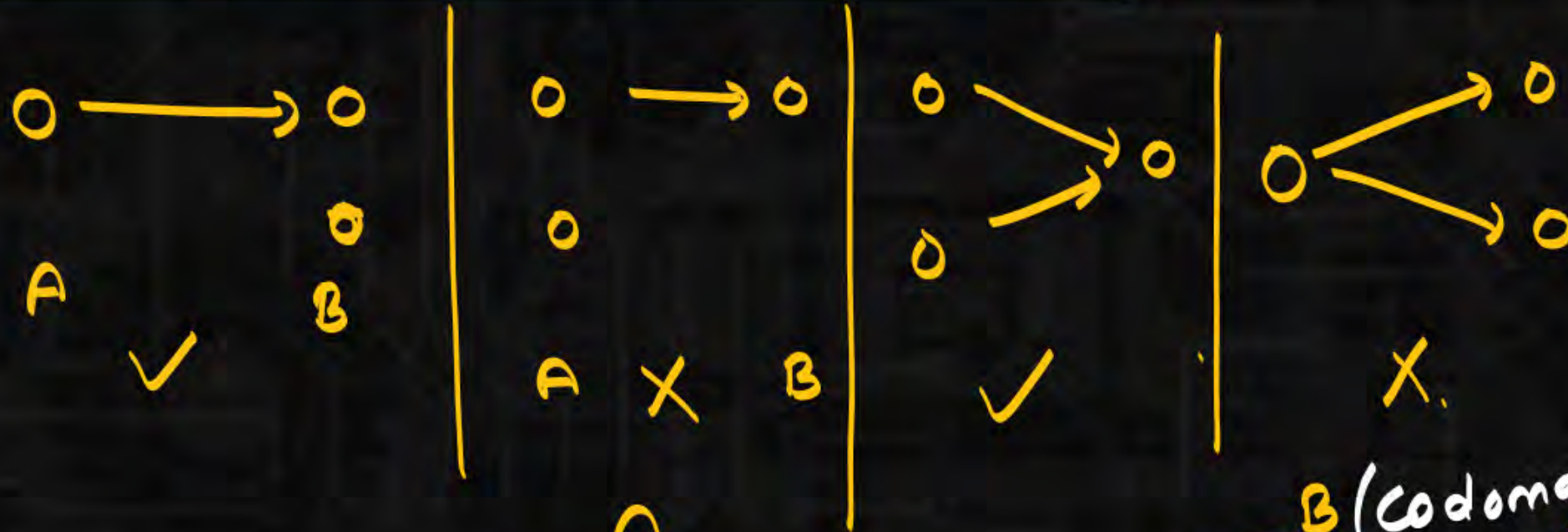
X



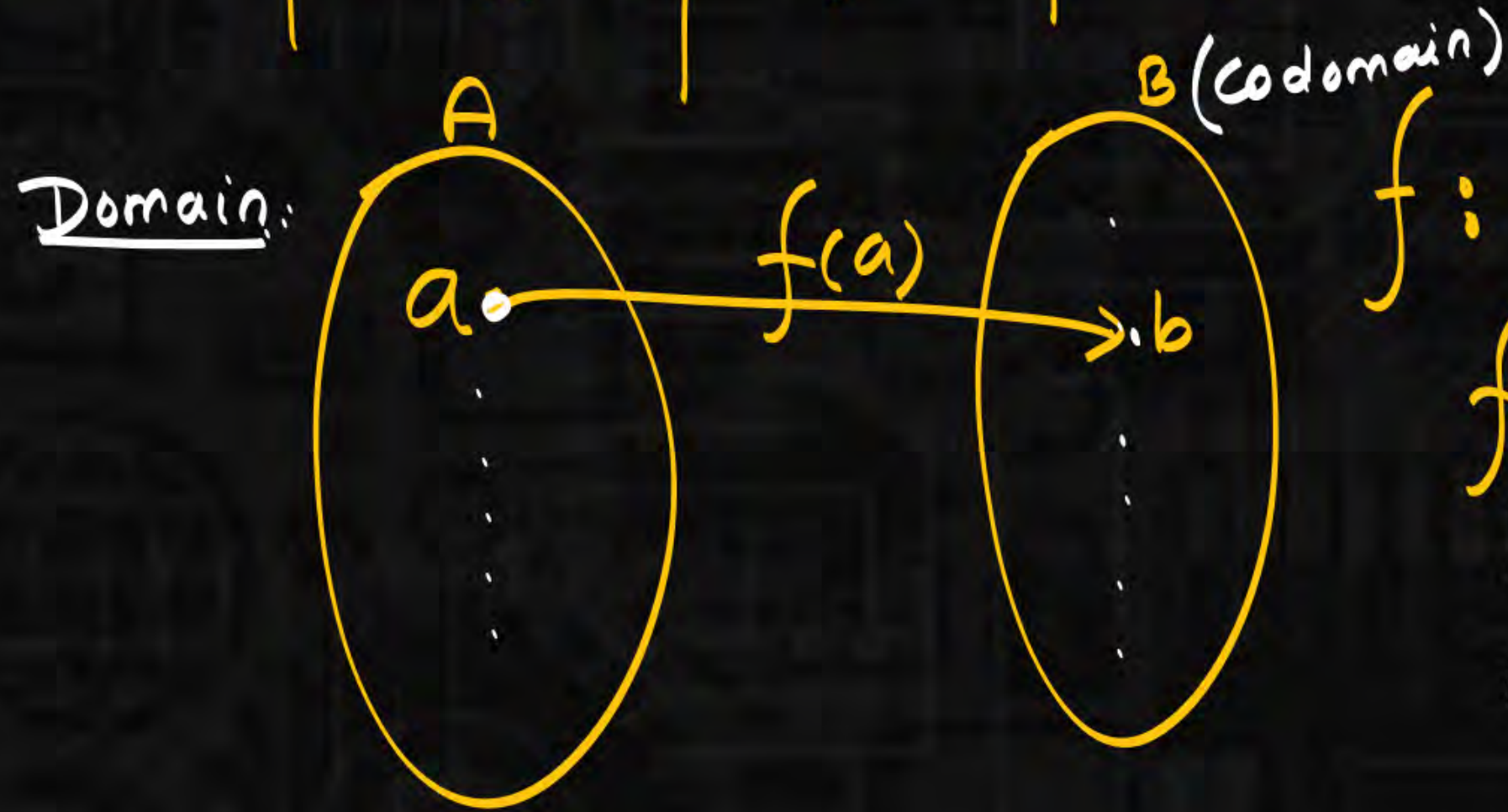
at same
time



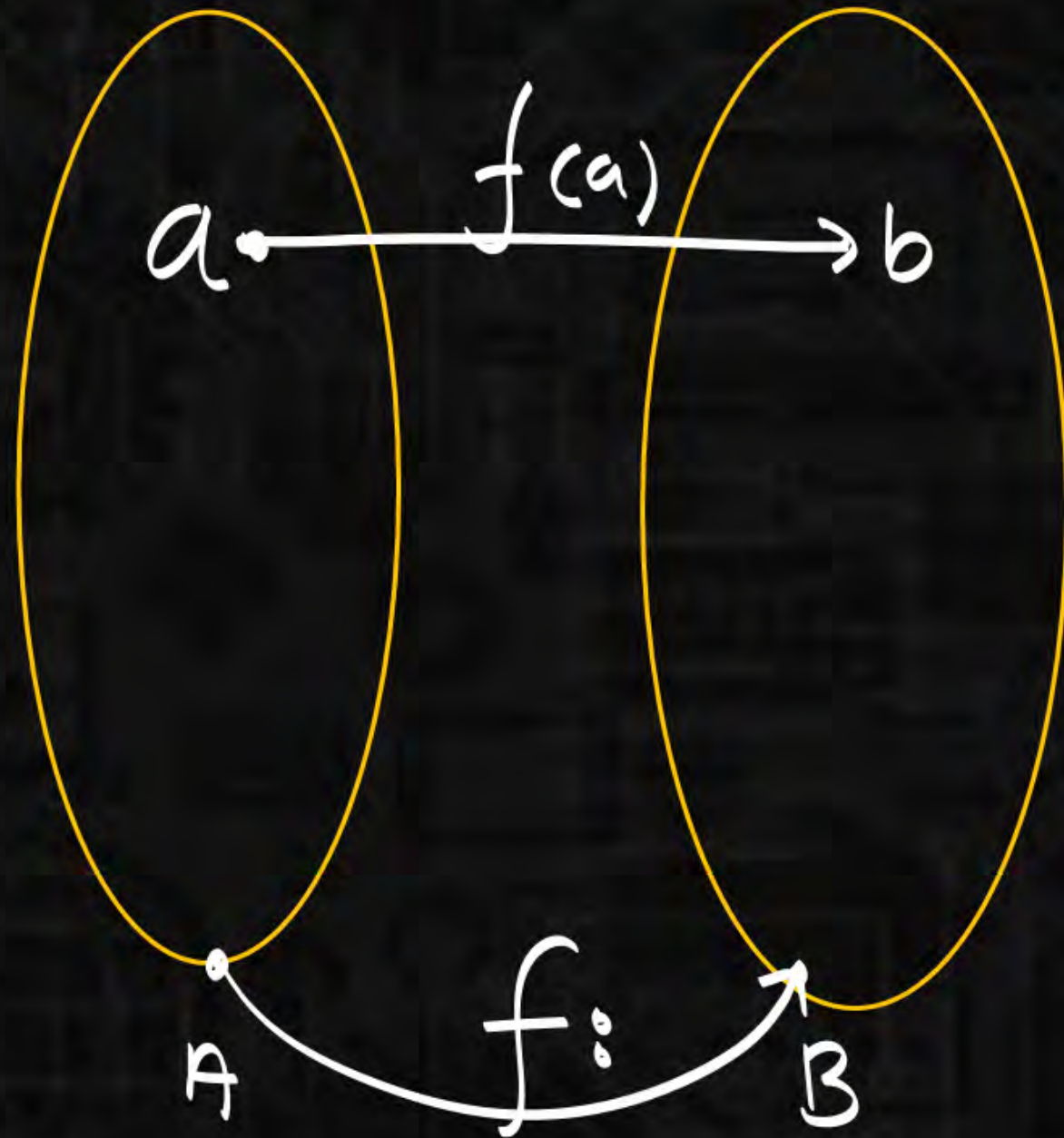
Functions



$$\begin{cases} f(x) = x^2 \\ f: \mathbb{Z} \rightarrow \mathbb{Z} \end{cases}$$

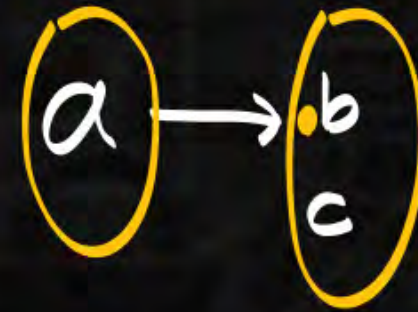


Functions



Range = set of images.

$\text{Range} \subseteq \text{codomain.}$



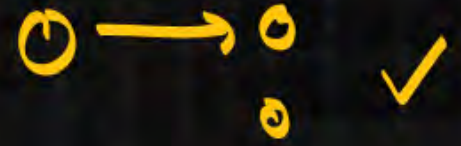
codomain: $\{b, c\}$

Range: $\{b\}$

Functions



$$f: A \rightarrow B$$



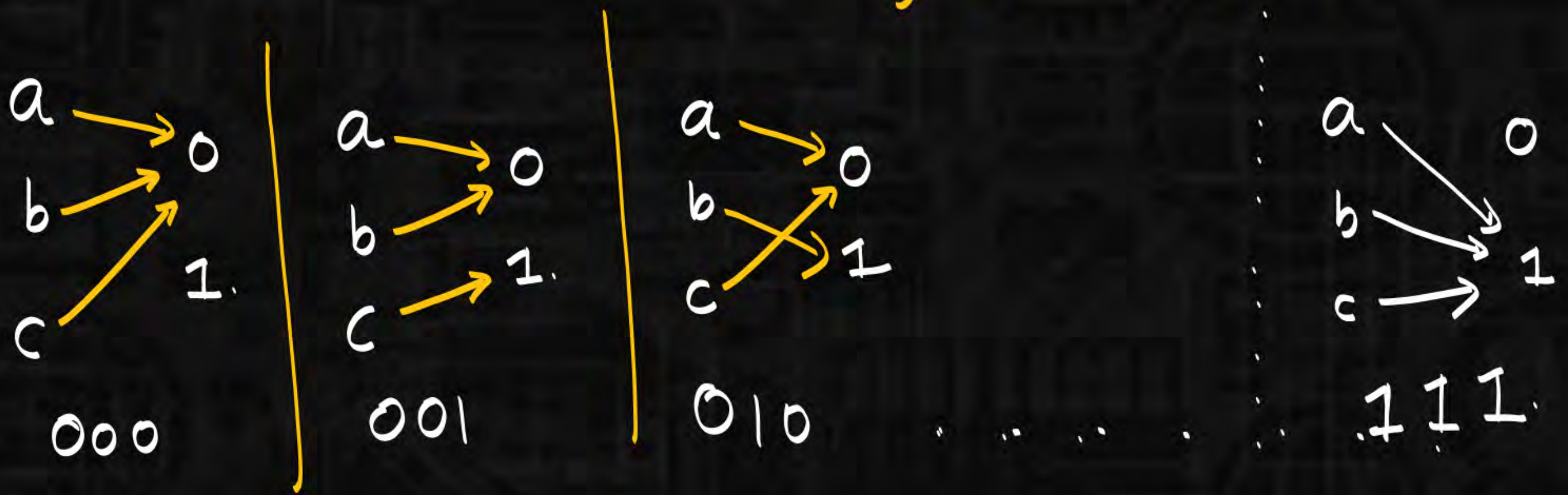
function is an assignment from A to B
such that all elements of A point to some
elements of B

one element of A can not point to at least 2 elements of B
@ same time

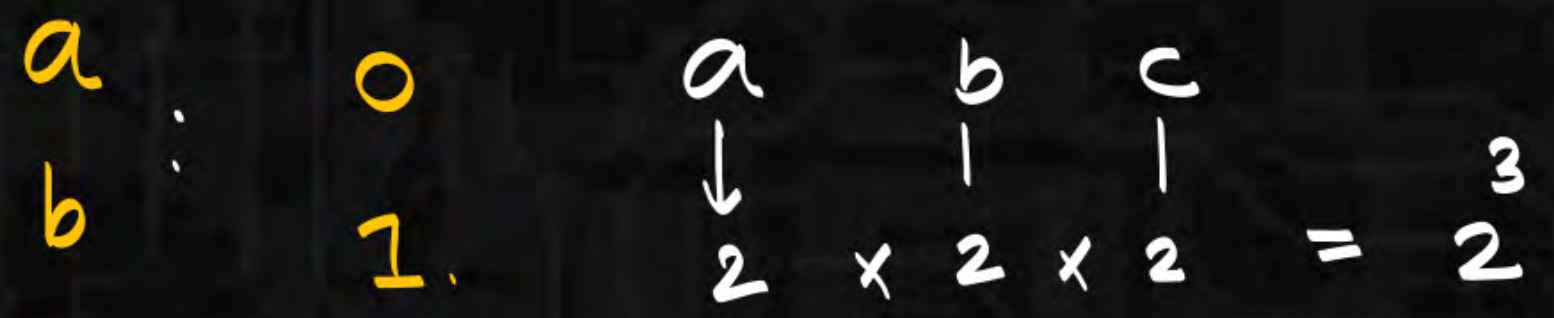
Functions

no. of diff. functions = no. of diff. arrows representation

$f: A \rightarrow B$ $|A|=3$ $|B|=2$ = no. of functions = 8



Functions



$$f: A \rightarrow B$$

$$|A| = 3 \quad |B| = 2$$

Total no. of functions

$= (R.S.)^{L.S.}$

Functions

$f: X \rightarrow Y$ Total no. of functions = $|Y|^{|X|}$ (GATE)

a) $|X| = 97$ $|Y| = 1$ $(1)^{97} = 1$

✓ b) $|X| = 1$ $|Y| = 97$ $(97)^1 = 97$

c) $|X| = 97$ $|Y| = 97$ $(97)^{97} = \dots$

d) not a.



Functions

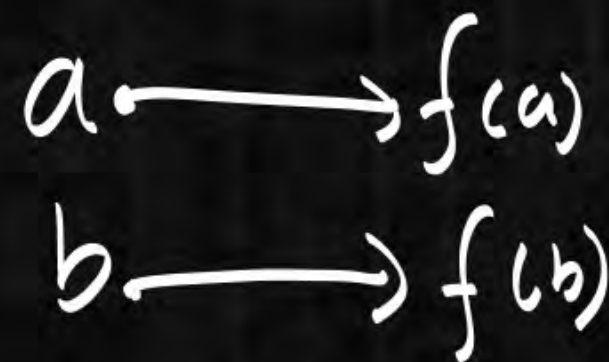
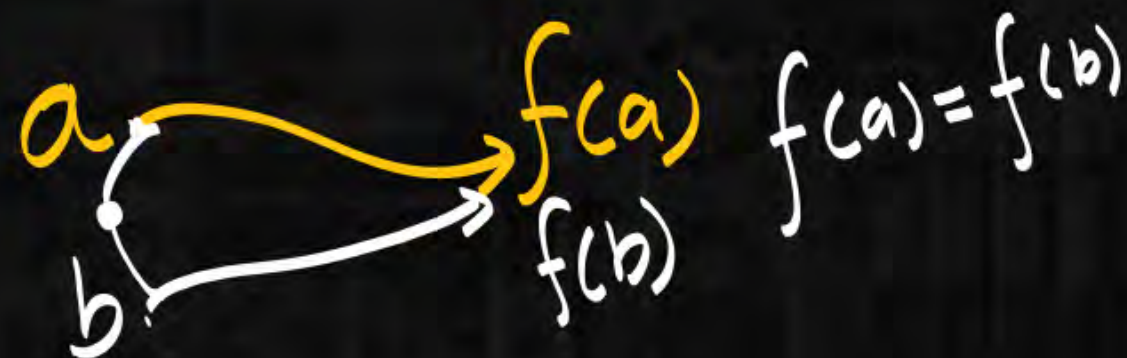


1:1 (one-to-one/injective)

f is called 1:1 function if it satisfies.

$$\forall a \forall b \left(f(a) = f(b) \rightarrow a = b \right) \text{ or } \left(a \neq b \rightarrow f(a) \neq f(b) \right)$$

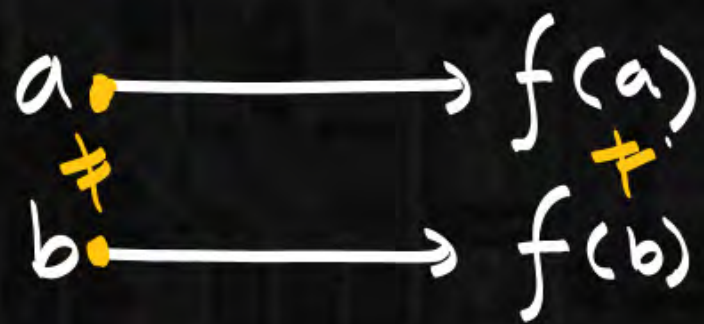
a, b are
elements in
domain.



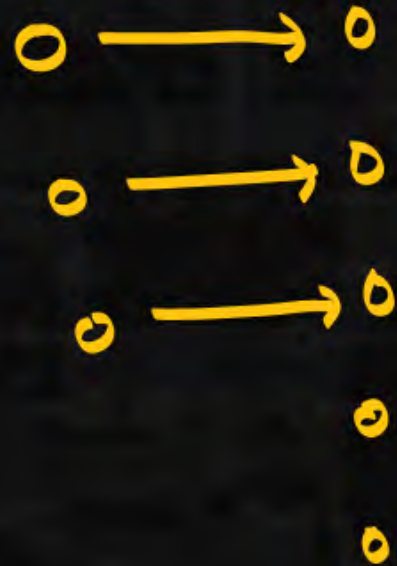
Functions



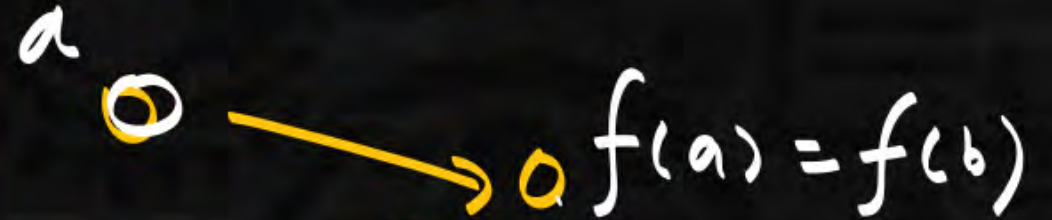
$$a \neq b \rightarrow f(a) \neq f(b)$$



valid:



Invalid:



$$f(a) = f(b) \rightarrow a = b$$

Functions

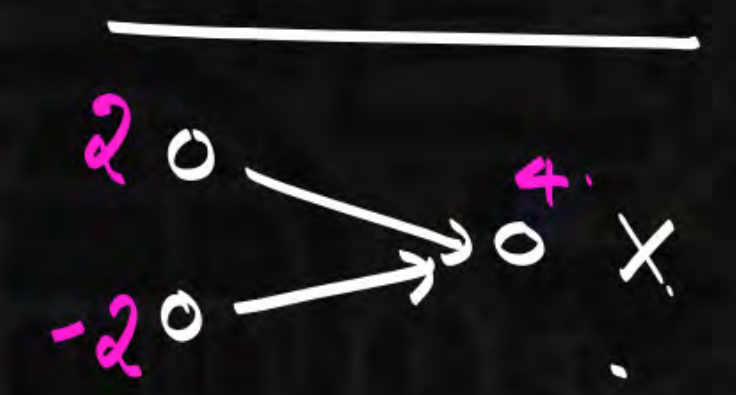
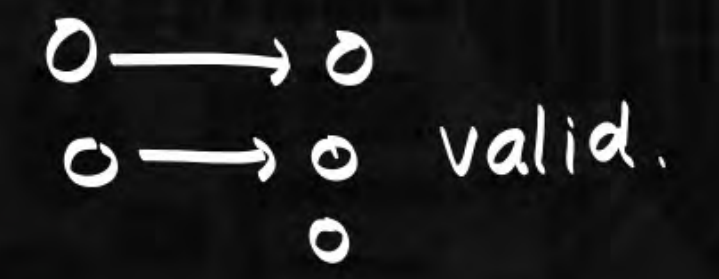
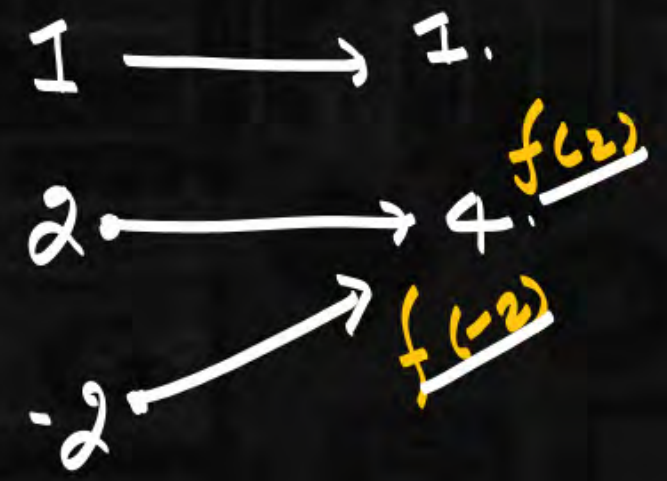
$$f(x) = x^2 \quad \text{not } 1:1$$

$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(a) = f(b) \rightarrow a = b$$

$$\underline{f(2) = f(-2) \rightarrow 2 = -2}$$

T.
F.



Functions



$$f: \underline{z} \rightarrow \underline{z}$$

$$f(x) = x + 1$$

it is 1:1 function

$$1 \longrightarrow 2$$

$$2 \longrightarrow 3$$

$$3 \longrightarrow 4$$

...



$$f(p) = f(w)$$

$$p + 1 = w + 1$$



$$p = w$$

$$f(a) = f(b)$$

$$\longrightarrow a = b$$

Functions

$f: A \rightarrow B$ (L.S > R.S 1:1 function is not possible)

(L.S ≤ R.S)
(A ≤ B)

A = 3

B = 5

5 ways

4 ways

3 ways



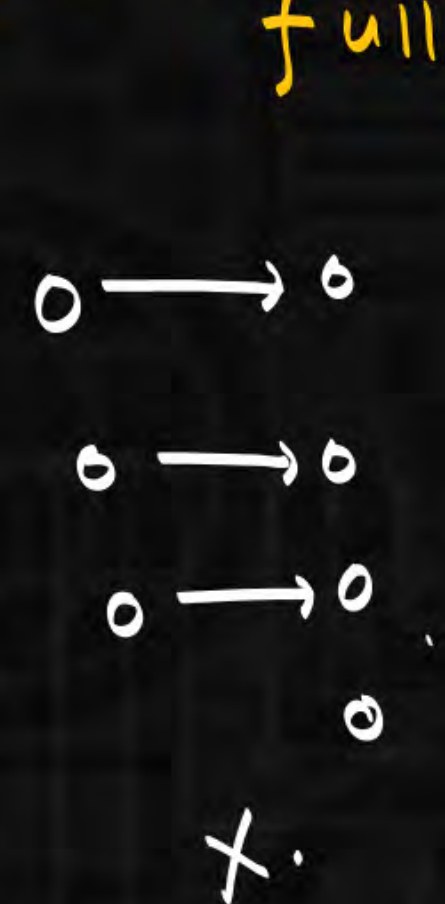
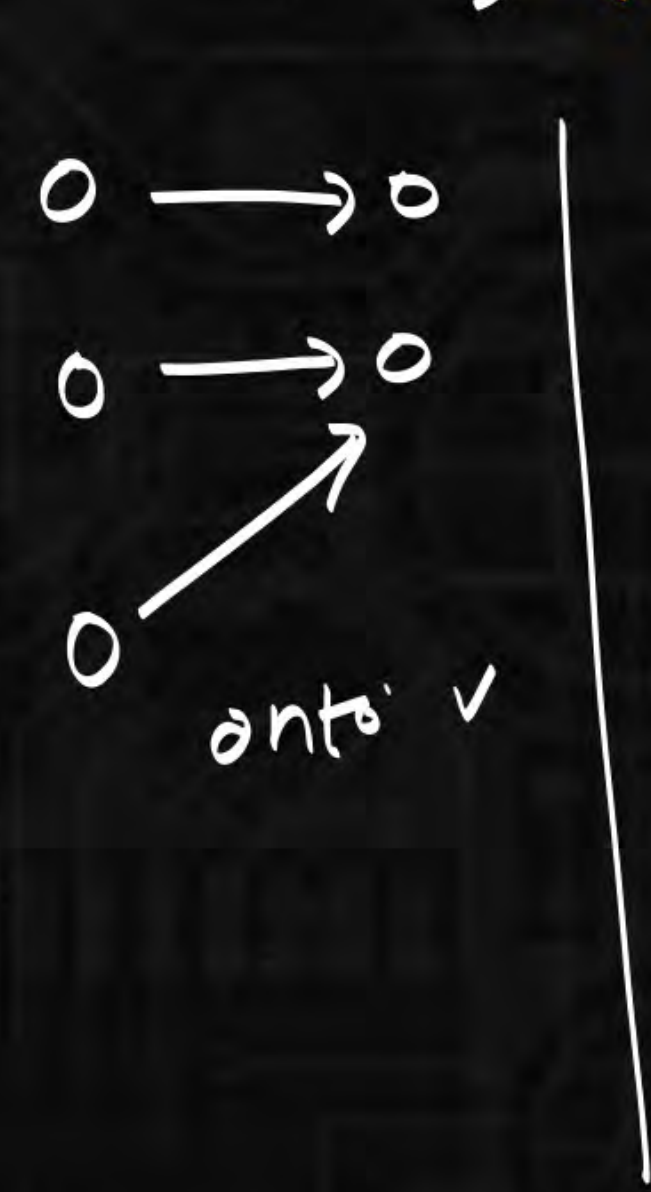
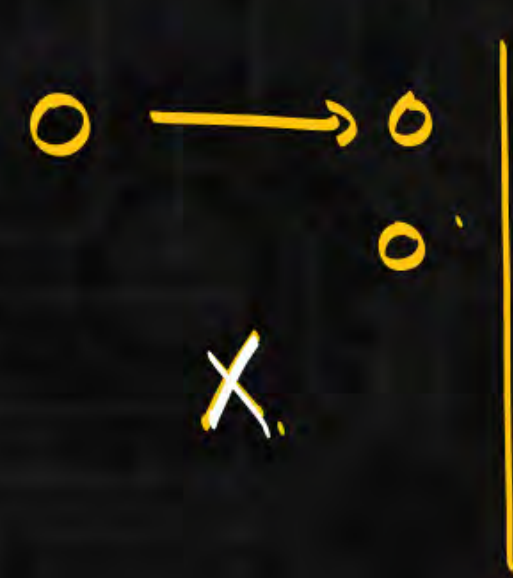
Total no. of 1:1 functions = $R.S P_{L.S}$

$$= 5 * 4 * 3 * \frac{2!}{2!}$$

$$= \frac{5!}{2!} = \frac{5!}{(5-3)!} = 5P_3$$

Functions

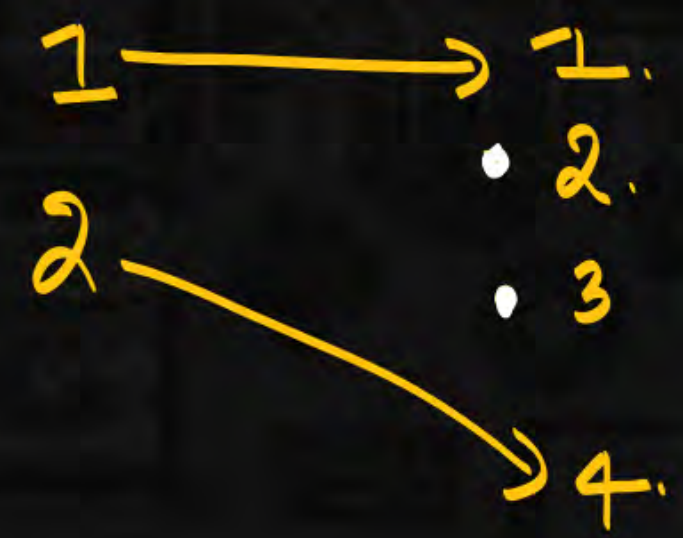
onto (surjective) (Range = codomain) (Right side is full)



Functions

$$f(x) = x^2$$

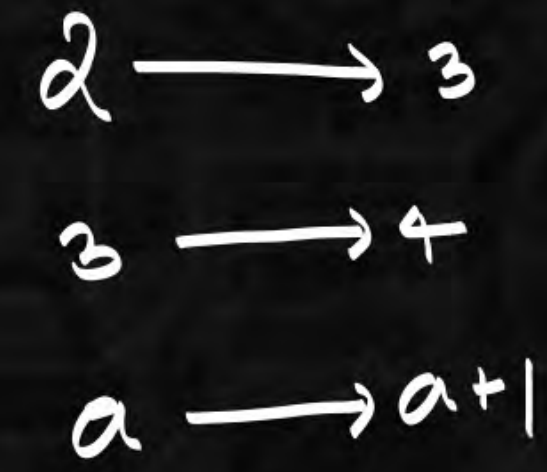
$$f: \mathbb{Z} \rightarrow \mathbb{Z} \quad \text{not onto}$$



$$f(x) = x + 1$$

$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

(Right side is full)



$$y = x + 1$$

$$y - 1 = \textcircled{x}$$



Functions



$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(x) = 2x - 3$$

x

y

$\bullet$

$\bullet 2$

$$y = 2x - 3$$

$$\frac{y+3}{2} = x \quad \text{we will not get}$$

$$y = 2$$

$$\frac{2+3}{2} = 5/2 \notin \mathbb{Z}$$



Functions

Total no. of onto functions:

$$f: A \rightarrow B \quad |A| \geq |B|$$

$$|A|=m \quad |B|=n \quad m \geq n$$

onto: (L.S < R.S)
 onto is not possible

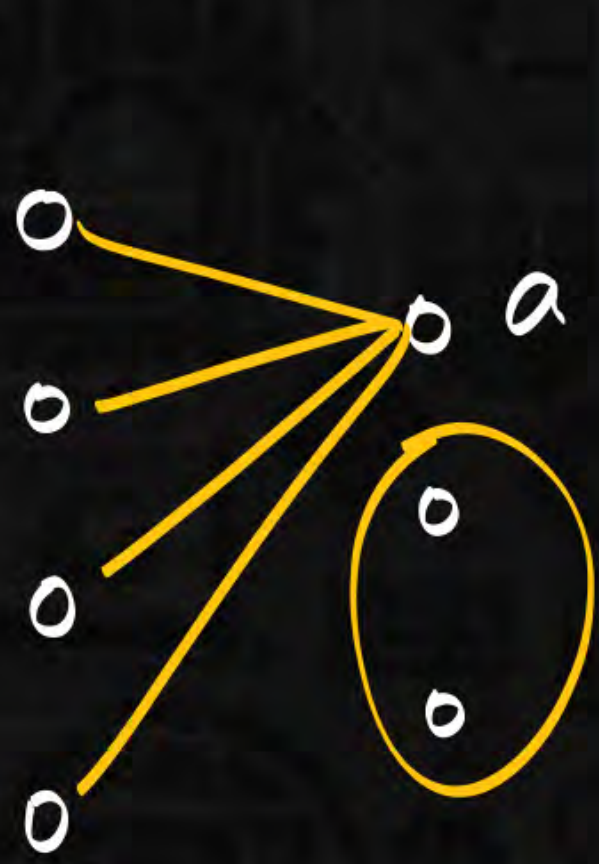
$$\begin{array}{c} 0 \rightarrow 0 \\ 0 \rightarrow 0 \\ 0 \end{array}$$

Total no. of functions = Total functions - non onto functions.

(L.S $\geq$ R.S)

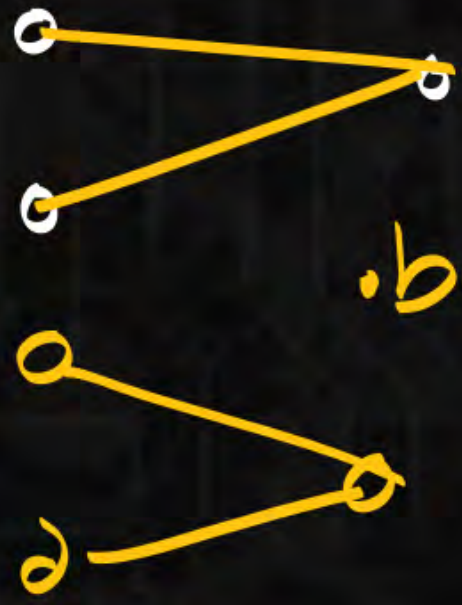
Functions

onto = Total functⁿ - non onto functⁿ.
 (Right side is not full)



Skipping 2 elements

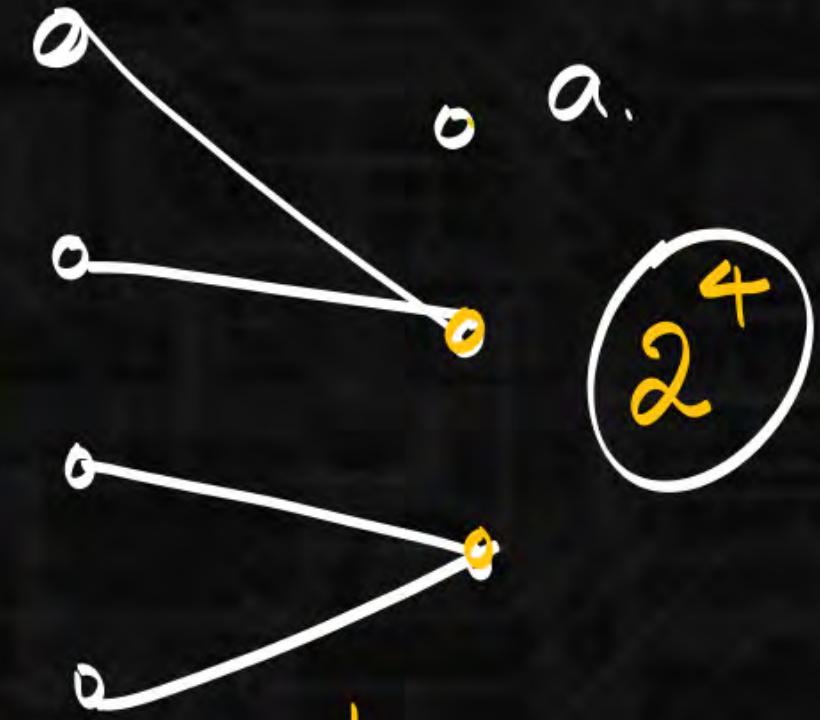
Functions



2^4

skip. b.

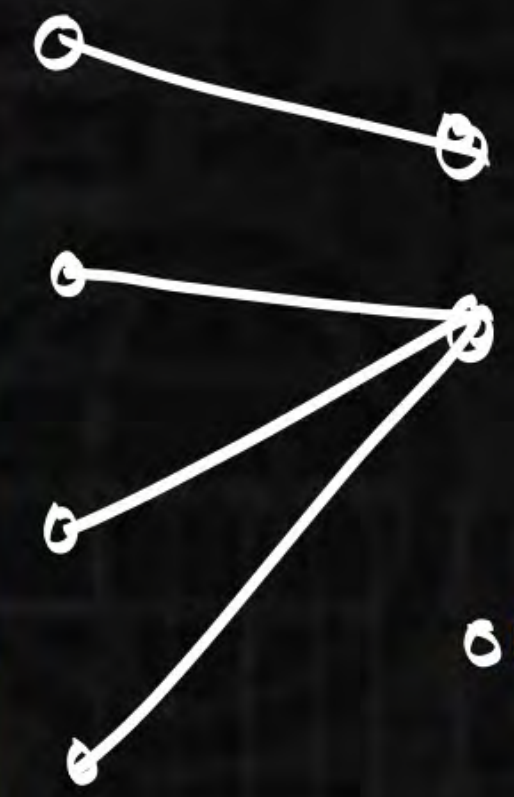
skip a.



a.

2^4

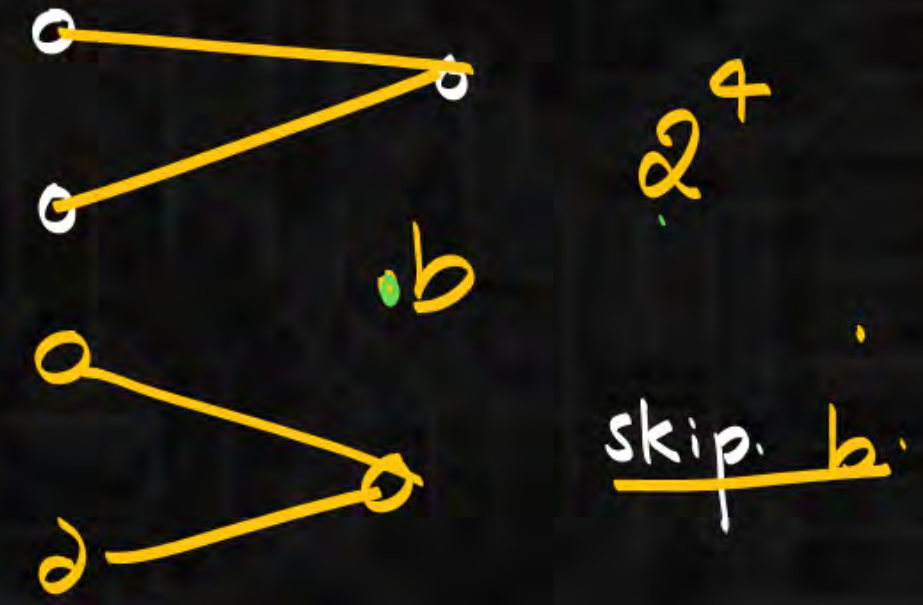
skip c.



2^4

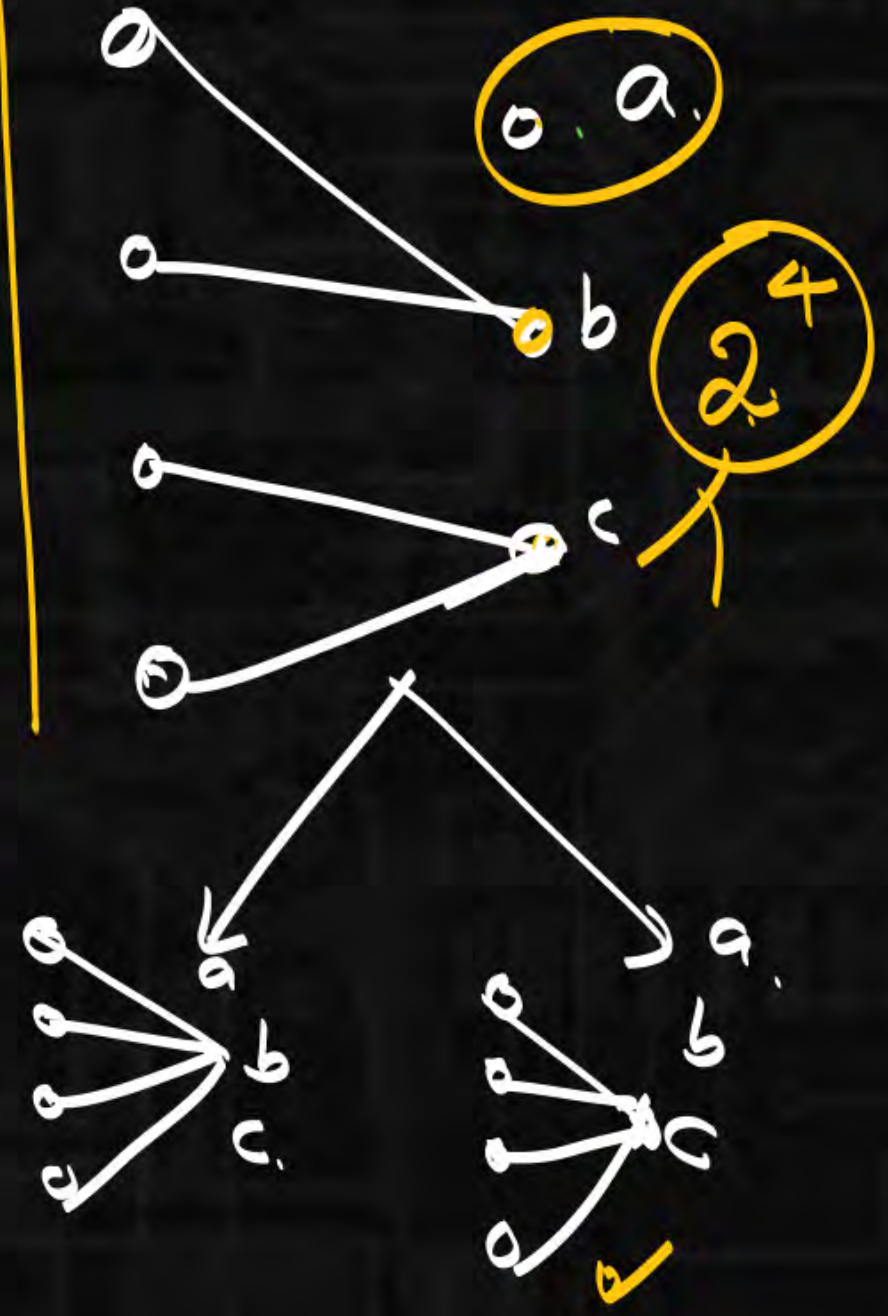
c

Functions

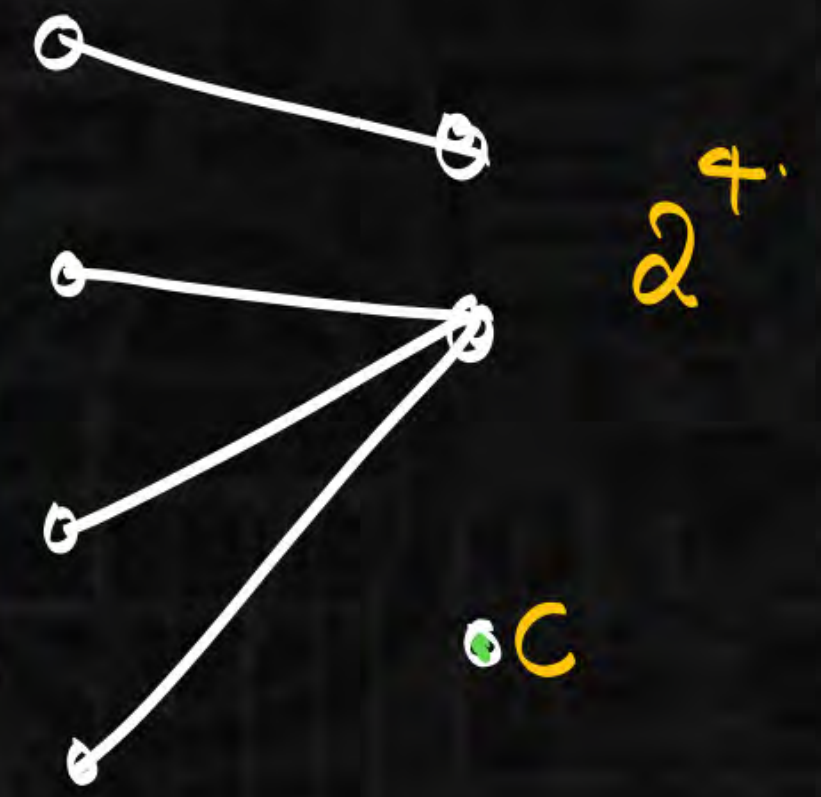


$$3c_1 \times 2^4 = 3 \cdot 2^4$$

skip a.



skip c.



Functions

a
b
c

0
0

Total functions = 2^3

2^3

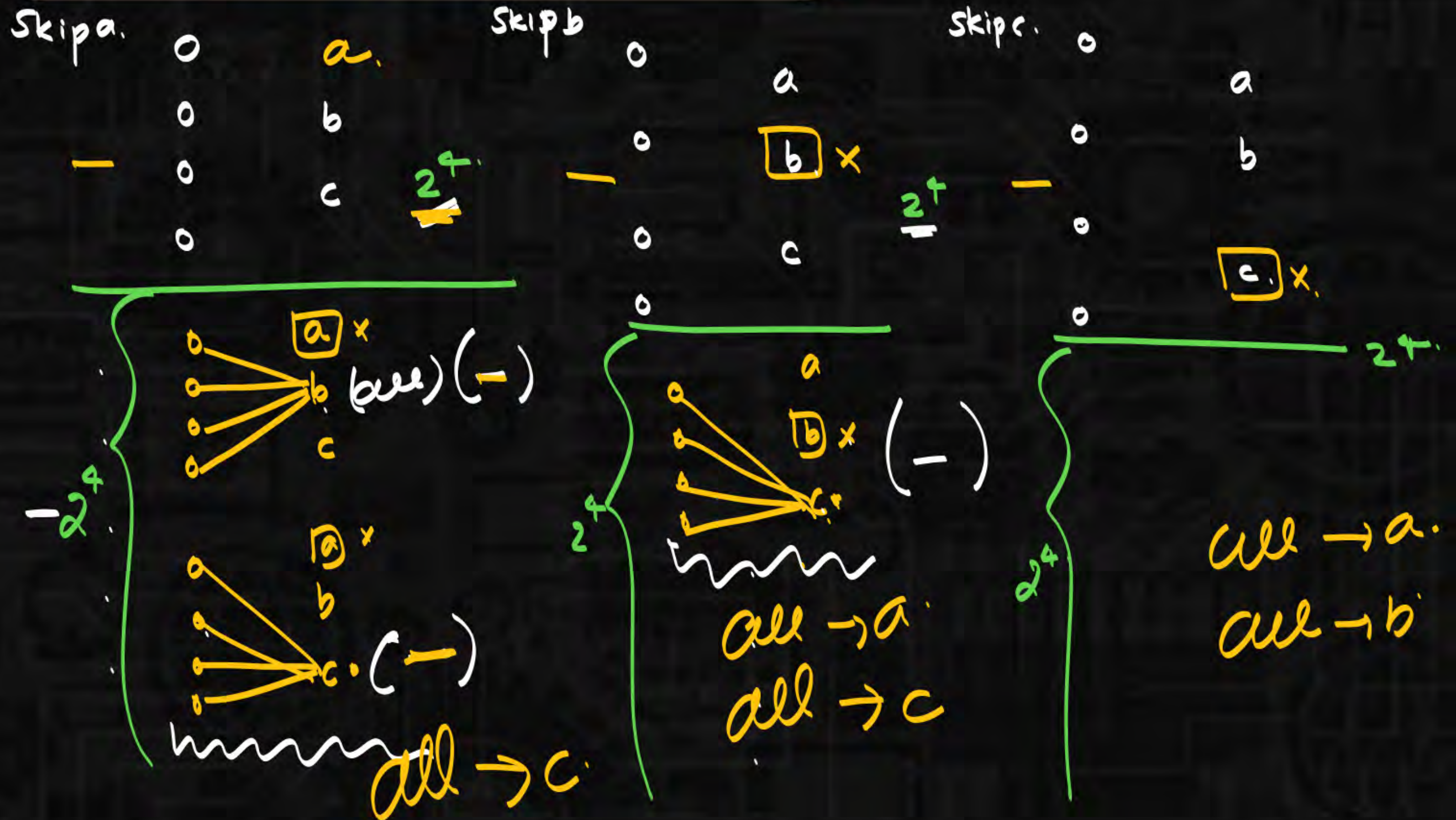


Functions

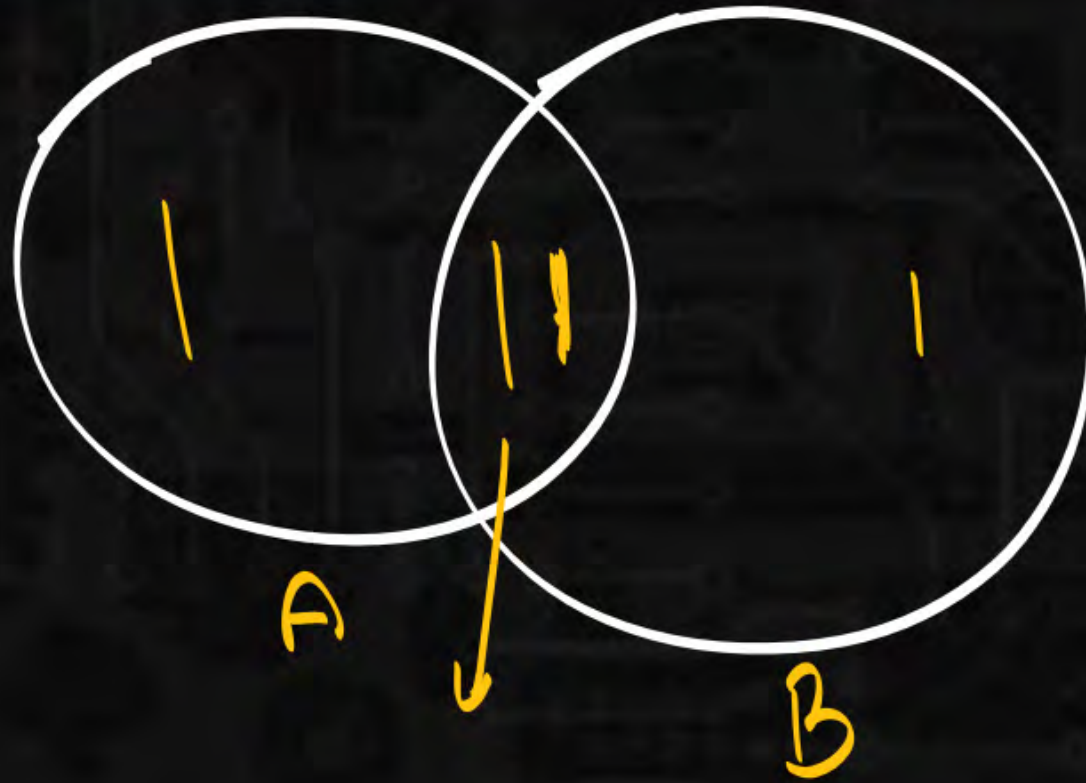
Total non onto = Total function - non onto functions. $|A| = 4$
 $|B| = 3$.

$$= 3^4 - \underbrace{3C_1}_{\text{ways to skip 1 element}} \cdot 2^4 + \underbrace{3C_2}_{\text{ways to skip 2 elements}} \cdot 1^4$$

Functions



Functions



$$A \cup B = A + B - (A \cap B)$$

Functions



$$A + B + C - A \cap B - B \cap C - A \cap C + A \cap B \cap C$$

Functions

$$f: A \rightarrow B \quad |A| = 4 \quad |B| = 3$$

Total onto functions = Total - total non onto.

$$= 3^4 - \underbrace{3C_1}_{\substack{\downarrow \\ \text{no. of} \\ \text{ways to} \\ \text{skip 1}}} 2^4 + \underbrace{3C_2}_{\substack{\downarrow \\ \text{no. of ways} \\ \text{to skip} \\ \text{2 elements}}}$$

Functions



$$f: A \rightarrow B \quad |A| = 4 \quad |B| = 3$$

$$A = m \quad B = n$$

$$n = 3$$

$$m = 4$$

$$\text{Total onto functions} = \text{Total} - \text{total non onto}$$

$$\sum_{i=0}^n (-1)^i \cdot n C_i \cdot (n-i)^m$$

$$= 3^4 - 3 C_1 2^4 + 3 C_2 1^4$$

$$= 3 C_0 (3-0)^4 - 3 C_1 (3-1)^4 + 3 C_2 (3-2)^4$$

$$= n C_0 (n-0)^m - n C_1 (n-1)^m + n C_2 (n-2)^m \dots$$

Functions

$$3c_0(3-0)^4 - 3c_1(3-1)^4 + 3c_2(3-2)^4 \dots$$

$$nc_0(n-0)^m - nc_1(n-1)^m + nc_2(n-2)^m$$

$$= \sum_{i=0}^n (-1)^i * nc_i * (n-i)^m$$

$$\begin{cases} m=7 & n=4. \\ (-1)^i n c_i (n-i)^m. \end{cases}$$

$$4c_0(4-0)^7 - 4c_1(4-1)^7 + 4c_2(4-2)^7 - 4c_3(4-3)^7 + 4c_4(4-4)^7$$

$$\underline{4^7 - 4 \times 3^7 - 6 \cdot 2^7 - 4 \cdot 1^7}$$

1:1 $f: \mathbb{Z} \rightarrow \mathbb{Z}$
onto.

$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(x) = x + 7$$

$$= 2x - 3$$

$$= -x + 5$$

$$= x^3$$

$$= x^2 + x$$

$$A = \{1, 2, 3, 4\}$$

$$B = \{1, 2, 3, 4, 5, 6\}$$

$f: A \rightarrow B$ how many 1:1 onto

$f: B \rightarrow A$ how many 1:1
onto.

